XIAMEN UNIVERSITY MALAYSIA
厦門大學馬來西亞分校

EEE312 Week 11
Lab Practice 2

Evaluate $\int_{-1}^{3} 3x^2 cos2x dx$

Using the Simpson's method with six subintervals.

Six subintervals between $x_0 = -1 \; to \; x_6 = 3$ means the distance h between subsequent points of x is

$$h = \frac{3 - 1(-1)}{6} = \frac{2}{3}$$

Calculate y for every point of x from $x_0 = -1 \; to \; x_6 = 3$ according to the table below.

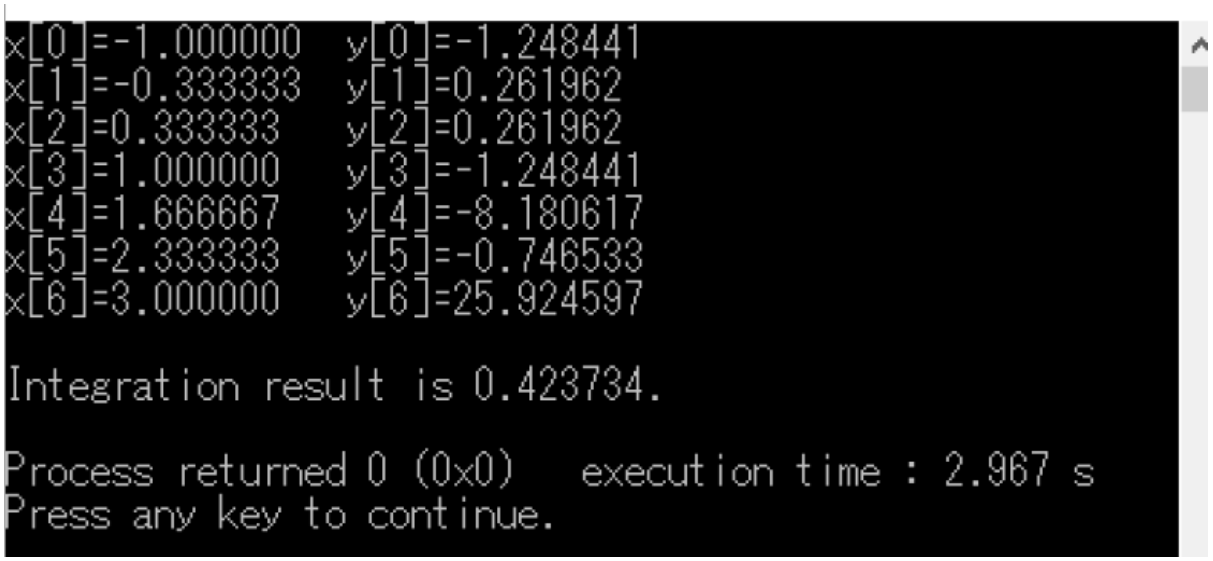
x	-1	?	?	?	?	?	3
y	?	?	?	?	?	?	?

Where $y = f(x) = 3x^2$cos2x

Implement the Simpson's equation in C code:

$$\int_{x_0}^{x_6} f(x)dx \approx \frac{h}{3}[y_0 + y_6 + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)]$$

Output


x[0]=-1.000000 y[0]=-1.248441
x[1]=-0.333333 y[1]=0.261962
x[2]=0.333333 y[2]=0.261962
x[3]=1.000000 y[3]=-1.248441
x[4]=1.666667 y[4]=-8.180617
x[5]=2.333333 y[5]=-0.746533
x[6]=3.000000 y[6]=25.924597
Integration result is 0.423734.
Process returned 0 (0x0) execution time : 2.967 s
Press any key to continue.